Classification of Stable Directed Refinement Systems

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1 Axiom System

Definition 1 (Stable Directed Refinement System)**.** *A* Stable Directed Refinement System *(SDRS) consists of a triple*

$$\mathcal{S} = (E, \preceq, D)$$

satisfying the following axioms.

(A1) **Directed completeness.** $(E, \preceq)$ is a directed-complete partial order.

(A2) **Scott continuity.** $D : E \to E$ is monotone and Scott-continuous.

(A3) **Algebraicity.** $(E, \preceq)$ is algebraic. Every element is the supremum of compact elements.

(A4) **Uniform local finite decomposition.** For every stabilised element e satisfying $D(e) = e$, there exists a finite set K_e of compact elements such that

$$e = \sup_{k \in K_e} k,$$

and there exists a uniform bound N with

$$|K_e| \leq N \quad \text{for all stabilised } e.$$

(A5) **Local generator sector.** For each stabilised element e, define

$$V_e = \mathrm{span}_{\mathbb{R}}(K_e).$$

Refinement directions act linearly and continuously on V_e.

(A6) **Incompatible refinement.** There exist refinement generators T_1, T_2 acting on some V_e such that

$$[T_1, T_2] \neq 0.$$

(A7) **Invariant quadratic structure.** Each V_e admits a positive-definite quadratic form

$$Q_e : V_e \to \mathbb{R}$$

preserved under continuous refinement flow.

(A8) **Generator regularity.** All refinement generators are continuous linear operators on each V_e.

(A9) **Compact width bound.** There exists $\ell > 0$ such that for every normalised stabilised state,

$$\Delta X \geq \ell$$

for the distinguished refinement direction X.

(A10) **Bounded symmetry algebra.** The stabilising symmetry algebra $\mathfrak{g}$ satisfies

$$\dim \mathfrak{g} \leq S < \infty$$

uniformly under refinement.

(A11) **Invariance–symmetry principle.** Any refinement-consistent invariant tangent direction lies in the symmetry-generated subspace.

(A12) **Non-integrability.** No unbounded sequence of independent conserved quantities exists modulo symmetry.

2 Stabilised Closure

Theorem 1. *Under (A1) and (A2), the refinement operator admits a least fixed point.*

Proof. The result follows from the Scott Fixed Point Theorem, since $(E, \preceq)$ is a dcpo and D is Scott-continuous. □

The stabilised sector is defined by

$$\mathcal{C} = \{e \in E \mid D(e) = e\}.$$

Proposition 1. *$\mathcal{C}$ inherits algebraicity from $(E, \preceq)$.*

Proof. Fixed points of Scott-continuous maps form a dcpo under inherited order. Algebraicity follows from (A3) together with finite decomposition condition (A4). □

3 Local Finite Generator Structure

For each stabilised element $e \in \mathcal{C}$, the associated generator sector V_e is finite-dimensional with

$$\dim V_e \leq N.$$

All subsequent structural derivations are carried out within an arbitrary but fixed V_e.

4 Lie Algebra Structure of Refinement Generators

Theorem 2. *Refinement generators acting on V_e form a Lie algebra under the commutator bracket.*

Proof. The commutator is bilinear and antisymmetric. The Jacobi identity follows from associativity of operator composition. □

5 Quadratic Invariance and Generator Structure

Fix a stabilised element $e \in \mathcal{C}$ and consider its finite-dimensional generator sector V_e.

The invariant quadratic form is written

$$Q_e(v) = \langle v, v \rangle,$$

where $\langle \cdot, \cdot \rangle$ denotes the associated symmetric bilinear form.

5.1 Antisymmetry of Generators

Continuous refinement flow generated by T satisfies

$$\frac{d}{dt}v(t) = Tv(t).$$

Proposition 2. *Preservation of Q_e under refinement flow implies*

$$\langle Tv, w\rangle = -\langle v, Tw\rangle \quad \textit{for all } v, w \in V_e.$$

Proof. Preservation of Q_e gives

$$\frac{d}{dt}\langle v(t), v(t)\rangle = 0.$$

Differentiation yields

$$\langle Tv, v\rangle + \langle v, Tv\rangle = 0,$$

which extends by bilinearity to all v, w. □

Thus refinement generators are antisymmetric with respect to $\langle\cdot,\cdot\rangle$.

5.2 Complex Structure

Theorem 3. *On each finite-dimensional V_e, antisymmetric generators relative to a positive-definite form induce a complex structure.*

Proof. Finite-dimensional antisymmetric operators admit block-diagonal decomposition into 2×2 rotational blocks. Each block admits an operator J satisfying

$$J^2 = -I.$$

The operator J extends over V_e by direct sum. □

Any antisymmetric operator on a finite-dimensional real inner-product space decomposes into rotational blocks together with a possible trivial one-dimensional kernel block.

5.3 Unitary Refinement Flow

Complexification

$$V_{e,\mathbb{C}} = V_e \otimes_{\mathbb{R}} \mathbb{C}$$

identifies multiplication by i with J.

Refinement flow therefore admits representation

$$\frac{d}{dt}\psi = -iH\psi,$$

with H symmetric.

Corollary 1. *Refinement evolution acts unitarily on $V_{e,\mathbb{C}}$.*

Proof. Symmetry of H implies $-iH$ generates a one-parameter unitary group. □

6 Uniqueness of the Quadratic Invariant Measure

The complexified sector $V_{e,\mathbb{C}}$ carries invariant inner product $\langle \cdot, \cdot \rangle$.

A weight functional

$$\mu : V_{e,\mathbb{C}} \to \mathbb{R}_{\geq 0}$$

is required to satisfy:

1. Non-negativity,
2. Orthogonal additivity,
3. Unitary invariance,
4. Continuity.

Theorem 4. *Under these conditions, μ is uniquely quadratic.*

Proof. Unitary invariance restricts μ to dependence upon component moduli. Orthogonal additivity produces the functional equation

$$f\left(\sqrt{x^2 + y^2}\right) = f(x) + f(y).$$

Continuity reduces this to the Cauchy equation on $\mathbb{R}_{\geq 0}$, whose unique solution is quadratic. □

Hence

$$\mu(\psi) = C\langle \psi, \psi \rangle.$$

After normalisation, $C = 1$.

7 Non-Commutativity and Minimal Dispersion States

Consider self-adjoint refinement generators X and P with non-vanishing commutator.

7.1 Uncertainty Inequality

Theorem 5. *For normalised ψ,*

$$\Delta X \, \Delta P \geq \frac{1}{2} \left| \langle [X, P] \rangle_\psi \right|.$$

Proof. The result follows from the Cauchy–Schwarz inequality applied to

$$(X - \langle X \rangle)\psi \quad \text{and} \quad (P - \langle P \rangle)\psi.$$

□

7.2 Characterisation of Minimal States

Theorem 6. *Equality in the uncertainty inequality holds if and only if there exists $\alpha \in \mathbb{R}$ such that*

$$(X + i\alpha P)\psi = \lambda\psi$$

for some complex scalar λ.

Proof. Equality in Cauchy–Schwarz requires linear dependence of the deviation vectors. Rearrangement yields the stated eigenvalue condition. □

8 Compact Width and Conditional Deformation Rigidity

The compact width bound (A9) asserts the existence of $\ell > 0$ such that

$$\Delta X \geq \ell$$

for all normalised stabilised states.

Theorem 7 (Conditional Deformation Rigidity)**.** *Assume the following:*

1. *The commutator $[X, P]$ depends analytically upon P.*
2. *Parity symmetry holds: $P \mapsto -P$ leaves the commutator invariant.*
3. *States saturating the uncertainty inequality exist.*
4. *The compact width bound is sharp.*

Then the leading deformation of the commutator is quadratic, and its coefficient is uniquely determined by ℓ.

Proof. Analyticity and parity symmetry restrict the leading correction to even powers of P. Thus

$$[X, P] = ic\left(1 + \gamma\ell^2 P^2 + O(P^4)\right).$$

Insertion into the uncertainty inequality yields

$$\Delta X \geq \frac{c}{2\Delta P} + \frac{c\gamma\ell^2}{2}\Delta P.$$

Minimisation under the assumption of saturating states produces

$$\Delta X_{\min} = c\sqrt{\gamma}\,\ell.$$

Sharpness of the compact width bound implies

$$\Delta X_{\min} = \ell,$$

which determines the coefficient. □

9 Instability Under Rank–Symmetry Imbalance

The symmetry algebra $\mathfrak{g}$ satisfies

$$\dim \mathfrak{g} \leq S < \infty$$

by (A10).

An interaction-supported subspace $I \subset V_e$ is assumed to arise under refinement.

A constraint map

$$\Phi : V_e \to \mathbb{R}^d$$

defines an invariant manifold

$$\mathcal{P} = \{v \in V_e : \Phi(v) = 0\}.$$

Interaction-induced constraint rank is defined by

$$r(v) = \operatorname{rank}\big(D\Phi(v)|_I\big).$$

Theorem 8 (Constraint–Symmetry Exclusion)**.** *If*

$$\inf_{v \in \mathcal{P}} r(v) \to \infty$$

under refinement while $\dim \mathfrak{g} \leq S$*, then no refinement-consistent invariant manifold persists at arbitrarily large refinement levels.*

Proof. The tangent space of $\mathcal{P}$ at v satisfies

$$T_v\mathcal{P} = \ker D\Phi(v).$$

Intersection with I yields

$$\dim(T_v\mathcal{P} \cap I) = \dim I - r(v).$$

As $r(v) \to \infty$, this dimension becomes strictly less than S.

By (A11), any invariant tangent direction must lie within the symmetry-generated subspace. Once

$$\dim(T_v\mathcal{P} \cap I) < S,$$

no nontrivial invariant directions remain.

Thus persistence fails. □

10 Classification Theorem

Theorem 9 (Classification of Stable Directed Refinement Systems)**.** *An SDRS satisfying axioms (A1)–(A12) admits precisely one of the following regimes:*

1. ***Stabilised representation regime.*** *Each local generator sector* V_e *carries:*
   - *(a) Lie algebra of refinement generators,*
   - *(b) Complex structure,*
   - *(c) Unitary refinement evolution,*
   - *(d) Unique quadratic invariant measure,*
   - *(e) Algebraically characterised minimal dispersion states,*
   - *(f) Quadratic commutator deformation under compact width.*
2. ***Instability regime.*** *If interaction-induced constraint rank diverges under bounded symmetry, no refinement-consistent invariant manifold persists.*

11 Hilbert Completion and Gaussian Form

Suppose the union of compact generator sectors is countable and locally finite under refinement.

The invariant quadratic form induces a norm

$$\|v\| = \sqrt{Q(v)}.$$

Theorem 10 (Separable Completion)**.** *Completion under this norm yields a separable Hilbert space. All structural results of the stabilised regime extend continuously.*

Proof. Completion under a positive-definite quadratic norm produces a Hilbert space. Continuity of refinement generators extends by density. □

Proposition 3 (Gaussian Form Under Canonical Representation)**.** *If a representation exists in which* X *acts by multiplication and* P *acts as the self-adjoint generator of translations, then minimal dispersion states assume Gaussian functional form.*

Axiom Dependency Summary

The structural results of this document depend upon the axiom system as follows. The existence of a least fixed point (Theorem 1) relies solely upon directed completeness (A1) and Scott continuity (A2). Algebraicity of the stabilised sector (Proposition 1) depends upon algebraicity of the base order (A3) together with uniform local finite decomposition (A4). Finite-dimensional generator sectors (Section 3) follow directly from (A4) and (A5). The Lie algebra structure of refinement generators (Theorem 2) uses (A5) and generator regularity (A8). Antisymmetry of generators (Proposition 2) and the induced complex structure (Theorem 3) depend upon invariance of the quadratic form (A7) together with finite dimensionality from (A4). Unitary refinement evolution (Corollary 1) follows from the complex structure and (A7). Uniqueness of the quadratic invariant measure (Theorem 4) relies upon unitary invariance derived from (A7) and continuity assumptions within Section 6. The uncertainty inequality (Theorem 5) depends upon non-commutativity guaranteed by incompatible refinement (A6). Characterisation of minimal dispersion states (Theorem 6) follows from Theorem 5 and Hilbert space structure. Conditional deformation rigidity (Theorem 7) requires the compact width bound (A9) together with the stated analytic and symmetry assumptions. The constraint–symmetry exclusion theorem (Theorem 8) relies essentially upon bounded symmetry dimension (A10) and the invariance–symmetry principle (A11), with non-integrability (A12) preventing trivial conservation escape. The final classification theorem (Theorem 9) is the direct aggregation of these dependencies.